

Lab

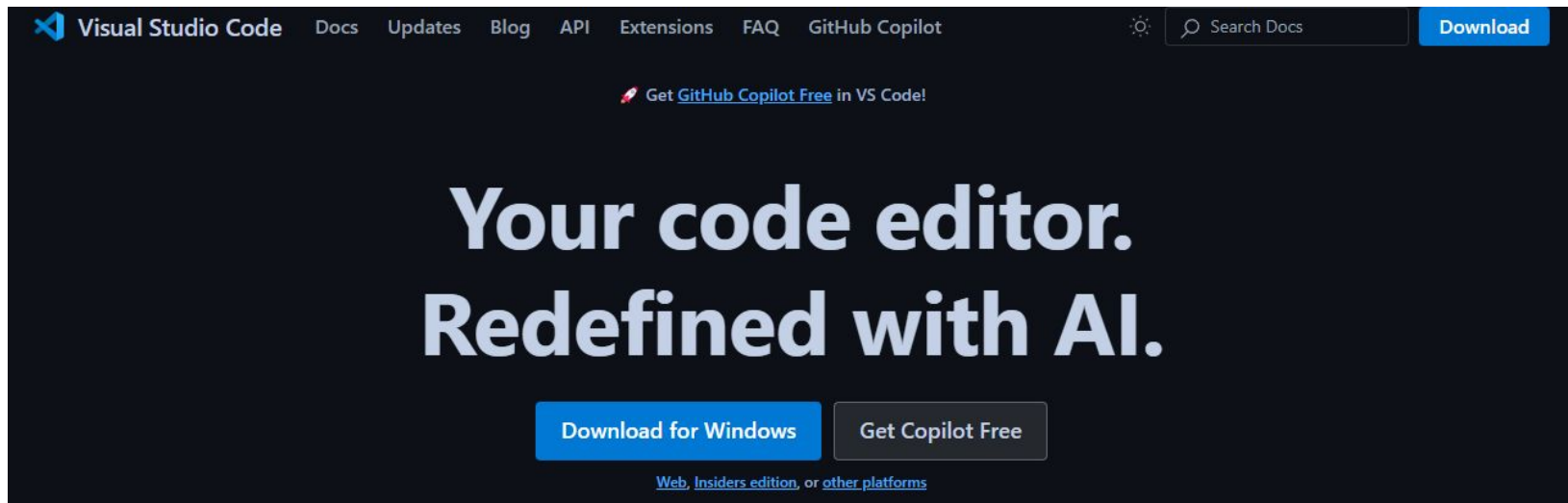
Lab Activities

1. Set up IDE - Visual Studio Code
2. Set up Git / GitHub
3. Set up Docker Container
 - a. DockerFile
 - b. DockerCompose
 - c. Requirements.txt
4. Run first py script
5. Run first notebook
6. Commit work

Lab

Set up IDE - Visual Studio Code

Download: <https://code.visualstudio.com/>

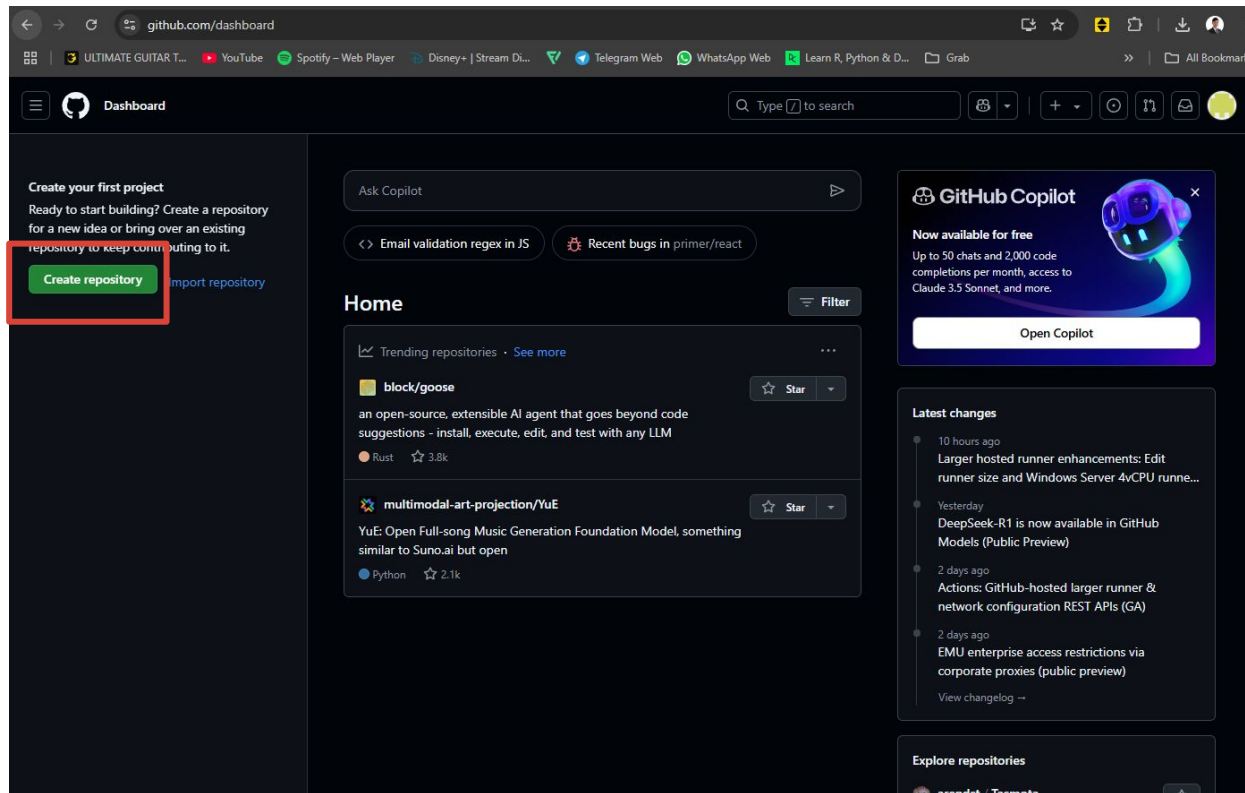


Lab

Set up Git / GitHub

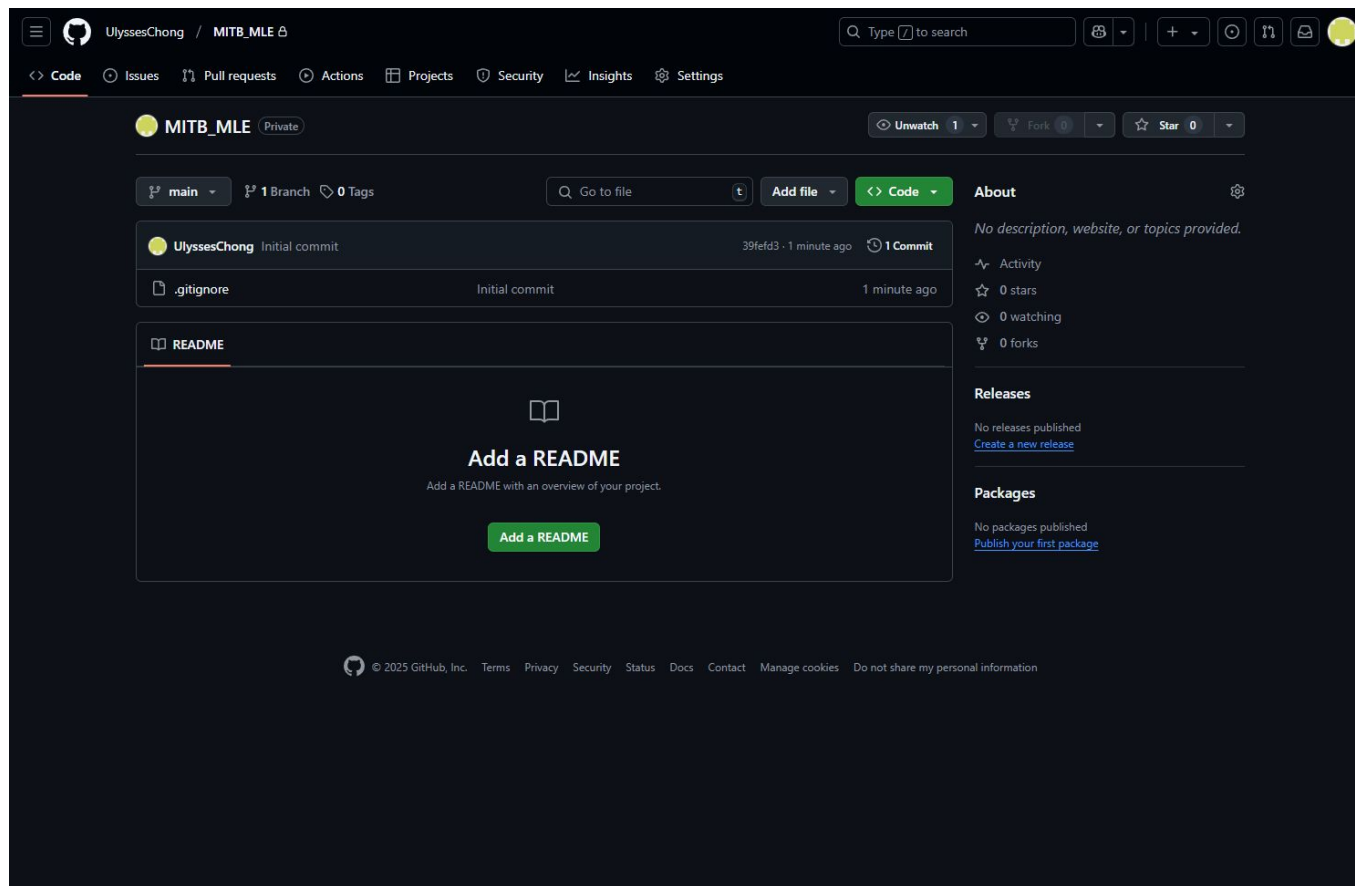
<https://github.com/>

highly recommend to enable 2FA



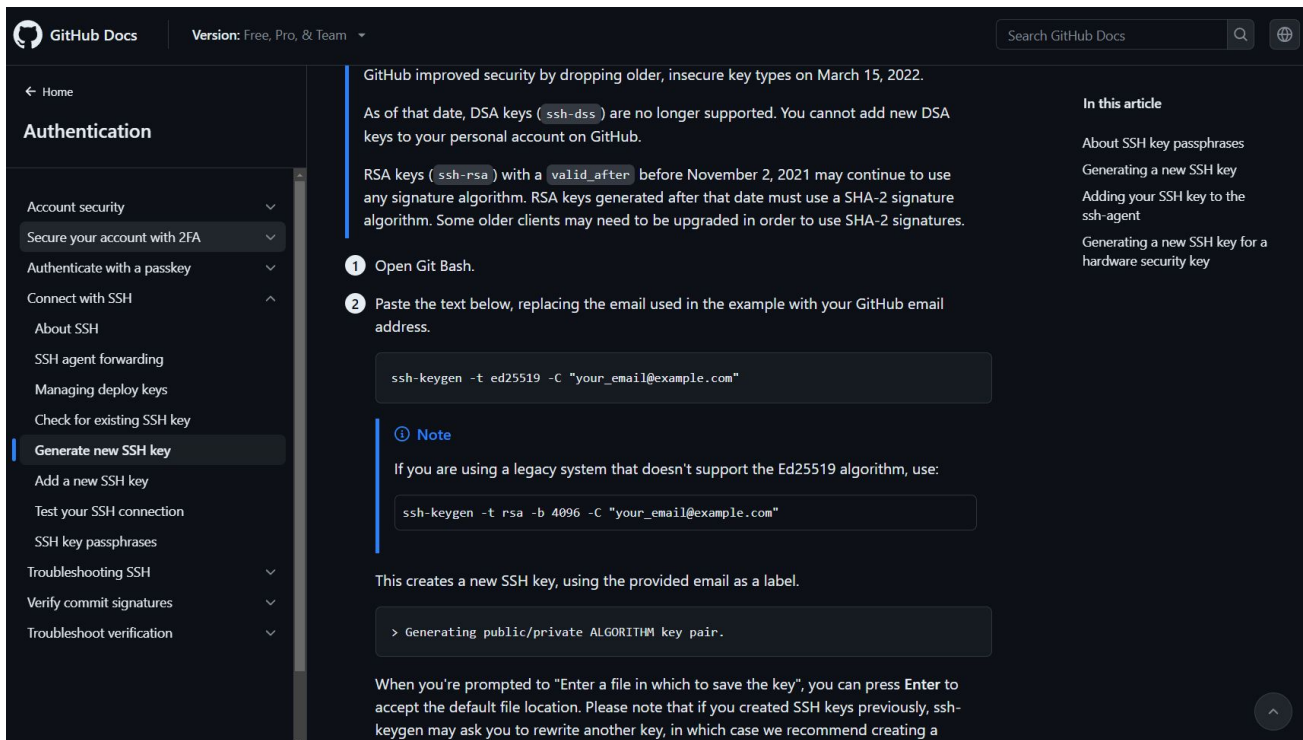
Lab

Create public repo Initialize .gitignore to Python



Set up SSH

<https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent>



The screenshot shows the GitHub Docs website. The left sidebar is open to the 'Authentication' section, with 'Generate new SSH key' selected. The main content area has a dark theme and contains the following text:

GitHub improved security by dropping older, insecure key types on March 15, 2022.

As of that date, DSA keys (`ssh-dss`) are no longer supported. You cannot add new DSA keys to your personal account on GitHub.

RSA keys (`ssh-rsa`) with a `valid_after` before November 2, 2021 may continue to use any signature algorithm. RSA keys generated after that date must use a SHA-2 signature algorithm. Some older clients may need to be upgraded in order to use SHA-2 signatures.

- 1 Open Git Bash.
- 2 Paste the text below, replacing the email used in the example with your GitHub email address.

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

Note

If you are using a legacy system that doesn't support the Ed25519 algorithm, use:

```
ssh-keygen -t rsa -b 4096 -C "your_email@example.com"
```

This creates a new SSH key, using the provided email as a label.

```
> Generating public/private ALGORITHM key pair.
```

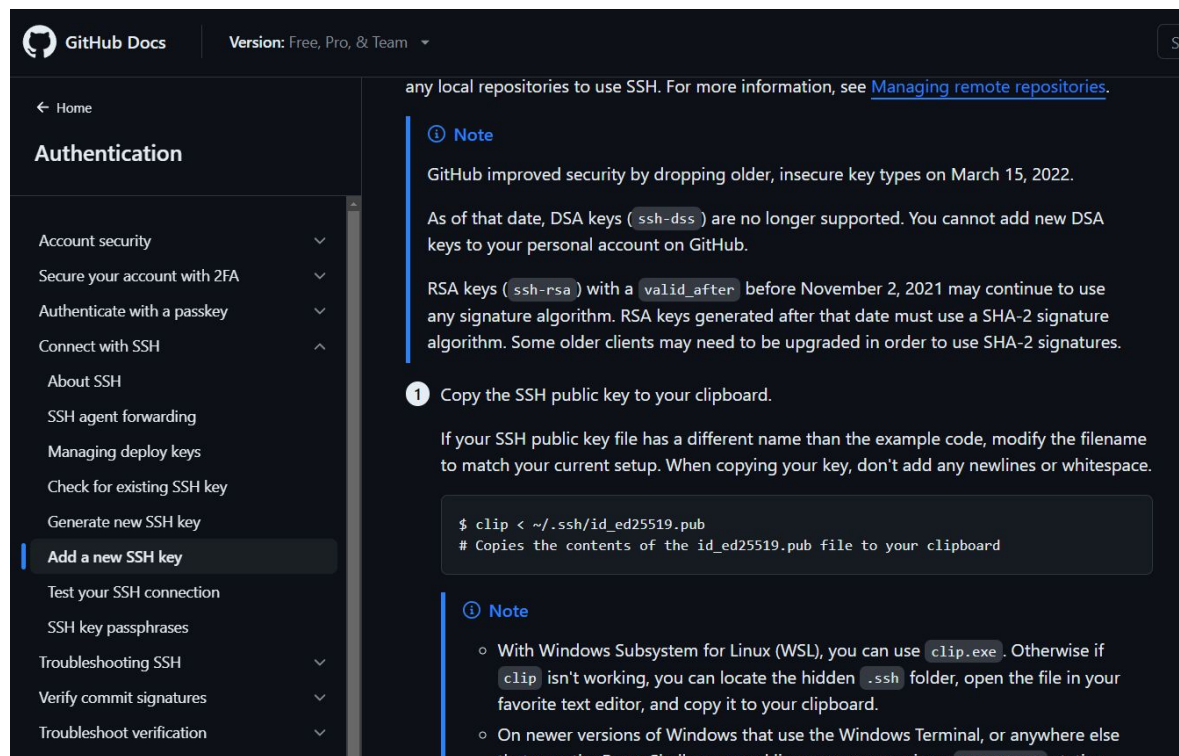
When you're prompted to "Enter a file in which to save the key", you can press Enter to accept the default file location. Please note that if you created SSH keys previously, `ssh-keygen` may ask you to rewrite another key, in which case we recommend creating a

In this article

- About SSH key passphrases
- Generating a new SSH key
- Adding your SSH key to the `ssh-agent`
- Generating a new SSH key for a hardware security key

Set up SSH

<https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account>



The screenshot shows the GitHub Docs website. The left sidebar is dark with a list of navigation items under the 'Authentication' section. The 'Add a new SSH key' item is highlighted with a blue bar. The main content area is light gray and contains a 'Note' section with text about GitHub's security improvements and SSH key support. Below the note, there is a numbered step '1' with instructions on how to copy the SSH public key to the clipboard, including a terminal command and a comment. Another 'Note' section at the bottom provides additional instructions for Windows users.

GitHub Docs Version: Free, Pro, & Team

← Home

Authentication

- Account security
- Secure your account with 2FA
- Authenticate with a passkey
- Connect with SSH
- About SSH
- SSH agent forwarding
- Managing deploy keys
- Check for existing SSH key
- Generate new SSH key
- Add a new SSH key**
- Test your SSH connection
- SSH key passphrases
- Troubleshooting SSH
- Verify commit signatures
- Troubleshoot verification

any local repositories to use SSH. For more information, see [Managing remote repositories](#).

Note

GitHub improved security by dropping older, insecure key types on March 15, 2022. As of that date, DSA keys (`ssh-dss`) are no longer supported. You cannot add new DSA keys to your personal account on GitHub.

RSA keys (`ssh-rsa`) with a `valid_after` before November 2, 2021 may continue to use any signature algorithm. RSA keys generated after that date must use a SHA-2 signature algorithm. Some older clients may need to be upgraded in order to use SHA-2 signatures.

1 Copy the SSH public key to your clipboard.

If your SSH public key file has a different name than the example code, modify the filename to match your current setup. When copying your key, don't add any newlines or whitespace.

```
$ clip < ~/.ssh/id_ed25519.pub
# Copies the contents of the id_ed25519.pub file to your clipboard
```

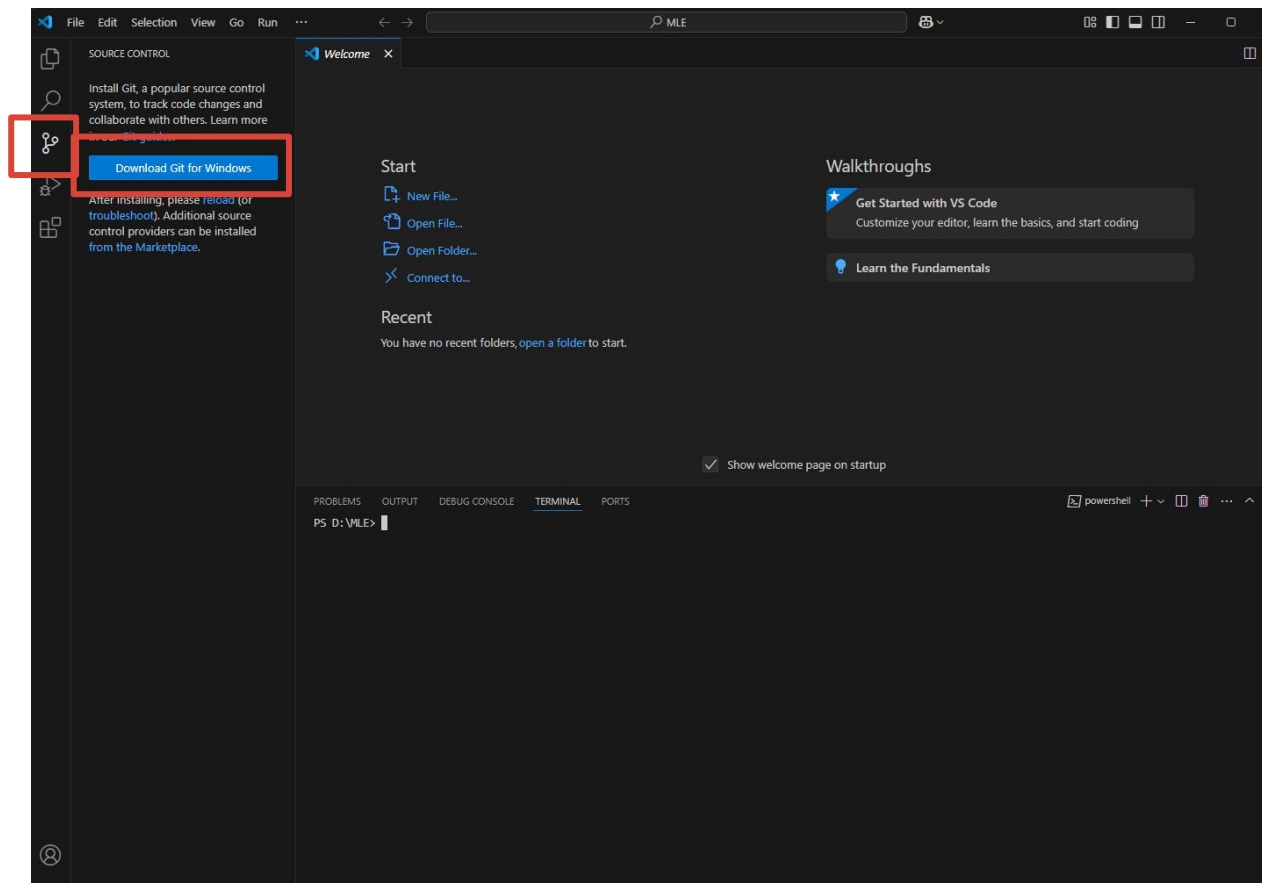
Note

- With Windows Subsystem for Linux (WSL), you can use `clip.exe`. Otherwise if `clip` isn't working, you can locate the hidden `.ssh` folder, open the file in your favorite text editor, and copy it to your clipboard.
- On newer versions of Windows that use the Windows Terminal, or anywhere else that uses the PowerShell command line, you may receive a `PowerPrompt` stating

Lab

set up git for windows

<https://git-scm.com/downloads/win>



Lab

clone your repo
git clone ...

Lab

quick git cmd:

git checkout main

git pull

git checkout -b "branch-1" # use any branch name with -
[make your changes to code]

git status

git add .

git commit -m "commit message"

git status

git push

Set up Docker Container

<https://www.docker.com/>

The screenshot shows the Docker website homepage. The main headline is "Develop faster. Run anywhere." with the subtext "Build with the #1 most-used developer tool". Below this, there are buttons for "Download Docker Desktop" and "Learn more about Docker". A dropdown menu is open under "Download Docker Desktop", showing options for Mac (Intel Chip, Apple Silicon), Windows (AMD64, ARM64 (BETA)), and Linux. The background features a large blue abstract graphic. At the bottom, there is a table of running containers and a cookie consent banner.

Name	Image	Status	CPU (%)	Ports	Actions
welcome-to-docker	docker/welcome-to-docker	Running	1.43%	52.52	Play Stop Delete
nginx	nginx:latest	Running	0%	80.80	Play Stop Delete

Lab

docker-compose build

docker-compose up

docker-compose down

Lab

Lab 1 completed sample

The screenshot shows a GitHub repository interface for the user UlyssesChong, specifically the MITB_MLE repository. The left sidebar displays the file explorer for the 'main' branch, with the 'lab_1' directory expanded. The files listed in 'lab_1' are DockerFile, docker-compose.yaml, hello.ipynb, hello.py, requirements.txt, .gitignore, and playground.py. The main content area shows the commit history for the 'lab_1' directory. A commit by UlyssesChong titled 'specify package versions' is highlighted. Below this, a table lists the files and their corresponding commit messages.

Name	Last commit message
..	
DockerFile	set up dockerfile
docker-compose.yaml	set up dockerfile
hello.ipynb	run docker
hello.py	run docker
requirements.txt	specify package versions

Are you feeling like a programmer
now?



An aerial night photograph of Singapore. The foreground shows a large, modern building with a brown roof and glass facade, identified by the 'SMU' logo. The background features a dense urban landscape with numerous illuminated skyscrapers under a dark sky. The text 'Questions?' is overlaid in a bold, orange font on the left side of the image.

Questions?

Thank You